

MURALI SAI BUDDAKKAGARI VENKATA

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SUMMARY

Applied AI Engineer specializing in LLM agent systems, multi-agent orchestration, and production eval pipelines. Shipped agentic workflows (LangGraph, LangChain, GPT-4o) including a Human-in-the-Loop system autonomously completing ~85% of tasks with full audit trail. Builds evaluation frameworks (RAGAS, faithfulness/relevancy metrics) to diagnose failure modes and improve agent solve rate through prompt and retrieval iteration. M.S. capstone: fused Faster R-CNN, Mask R-CNN, and YOLO26 into a size-conditioned ensemble for medical-imaging cell detection, outperforming every individual model at 0.71 mAP@0.5.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, PySpark, TypeScript
- **AI & Machine Learning:** Prompt Engineering & Design, Agent Evaluation (RAGAS), Retrieval-Augmented Generation, Failure Mode Analysis, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, Transformers, HuggingFace, PyTorch, TensorFlow, Model Context Protocol (MCP), FastAPI, ChromaDB, Scikit-learn, Embedding Models, LangSmith, LLMOps
- **Infrastructure & DevOps:** AWS (ECS Fargate, S3), Google Cloud Platform, BigQuery, Azure, Docker, PostgreSQL, GitHub Actions, CI/CD, ReactJS, OpenAI API, Claude API

EXPERIENCE

Querkey Inc

Jun 2025 - Aug 2025

Data Science Intern

CA, US

- Engineered end-to-end ML data ingestion pipelines (Python, PySpark, GCP, BigQuery) processing 200+ MP4 files daily, cutting validation latency by 40% and eliminating manual bottlenecks across 3 production data streams.
- Reduced production ML workflow failure rate by 60% by deploying a Dockerized observability stack (MLflow, HuggingFace monitoring, ChromaDB, FastAPI) over 100K+ records, enabling real-time failure detection and proactive incident response for core pipelines.

VIEW Synergy

Jan 2024 - Apr 2024

AI and Automation Intern

KA, IND

- Reviewed React, and MySQL layers across the live codebase and traced 2 production exceptions in Azure Functions to their root cause, in a technical assessment that documented 94 code level and security issues, including critical SQL injection vulnerabilities.
- Documented Microsoft Entra ID app registration and role setup guides and audited 6 backend microservices for legacy Redis usage, supporting a proposed Azure hosting migration projected to save \$51.59 per month (\$619 annually).

PROJECTS

SEC EDGAR RAG Filing Analyzer

- Deployed a production RAG system answering compliance-sensitive questions over 8,232 SEC 10-K filing chunks, verified by 475 automated tests and a 54-question eval suite (0.853 faithfulness, 0.936 citation accuracy, 0.981 refusal correctness) at \$0.00004/test and under \$1/month, by building a FastAPI + ChromaDB pipeline with cross-encoder reranking, RBAC information barriers (FINRA Rule 2241 / SEC Rule 15g-1), and prompt-injection guardrails.
- Raised answer faithfulness 0.697 to 0.853 (2.5x beyond measurement noise) and eliminated cross-company answer contamination, by diagnosing and fixing three retrieval bugs, including a mis-scoped confidence gate and an entity-detection gap letting Goldman Sachs questions retrieve Tesla's filings, confirmed across three independent test runs.

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Shipped a production multi-agent system achieving an 85% autonomous task-completion rate across long-horizon workflows, validated by 106/113 passing tests against a live deployment, by architecting a 3-layer LangGraph hierarchy (supervisor, parallel specialists, reviewer) with GPT-4o/Claude Haiku model routing and two-tier Redis + pgvector memory.
- Closed 12 production-blocking defects (7 found in a formal spec audit, 5 the audit missed), restoring an end-to-end review-rework-escalate loop and a Human-in-the-Loop system routing 15% of edge cases with full audit trail, by tracing a routing bug that had silently made the system's revision path unreachable since launch.

PUBLICATIONS AND CERTIFICATIONS

- Adaptive Ensemble Detection for Medical Imaging, M.S. Capstone, Stevens Institute of Technology (Advised by Prof. Kevin Lu). A size-conditioned Weighted Box Fusion ensemble of Faster R-CNN, Mask R-CNN, and YOLO26 for cell-nucleus detection, reaching 0.71 mAP@0.5 on the 2018 Data Science Bowl benchmark.
- Best Research Paper Award, ICICC 2024 - WorkFit: Real-time Multimodal Posture Monitoring System using CNN-based Computer Vision and Flex Sensor Fusion.

EDUCATION

Stevens Institute of Technology

May 2026

Master of Science, Applied Artificial Intelligence

NJ, US

PES University

May 2024

Bachelor of Technology, Computer Science (Specialization in ML)

KA, IND